

# Project Profile — MiniMax H3 Local Multimodal Video Studio on AMD Radeon

AMD AI DevMaster Hackathon 2026 · Track 1: Development of Multimodal Content Creation Tools

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## 1. Project Background

Generative video models are becoming central to content creation — advertising, short-form social video, game cinematics, e-learning and IP asset pipelines. MiniMax H3 is an **open-weights, omni-modal** model that turns text and/or images into video with **synchronized audio**, covering text-to-video (T2V), image-to-video (I2V) and reference-to-video (R2V) in one architecture.

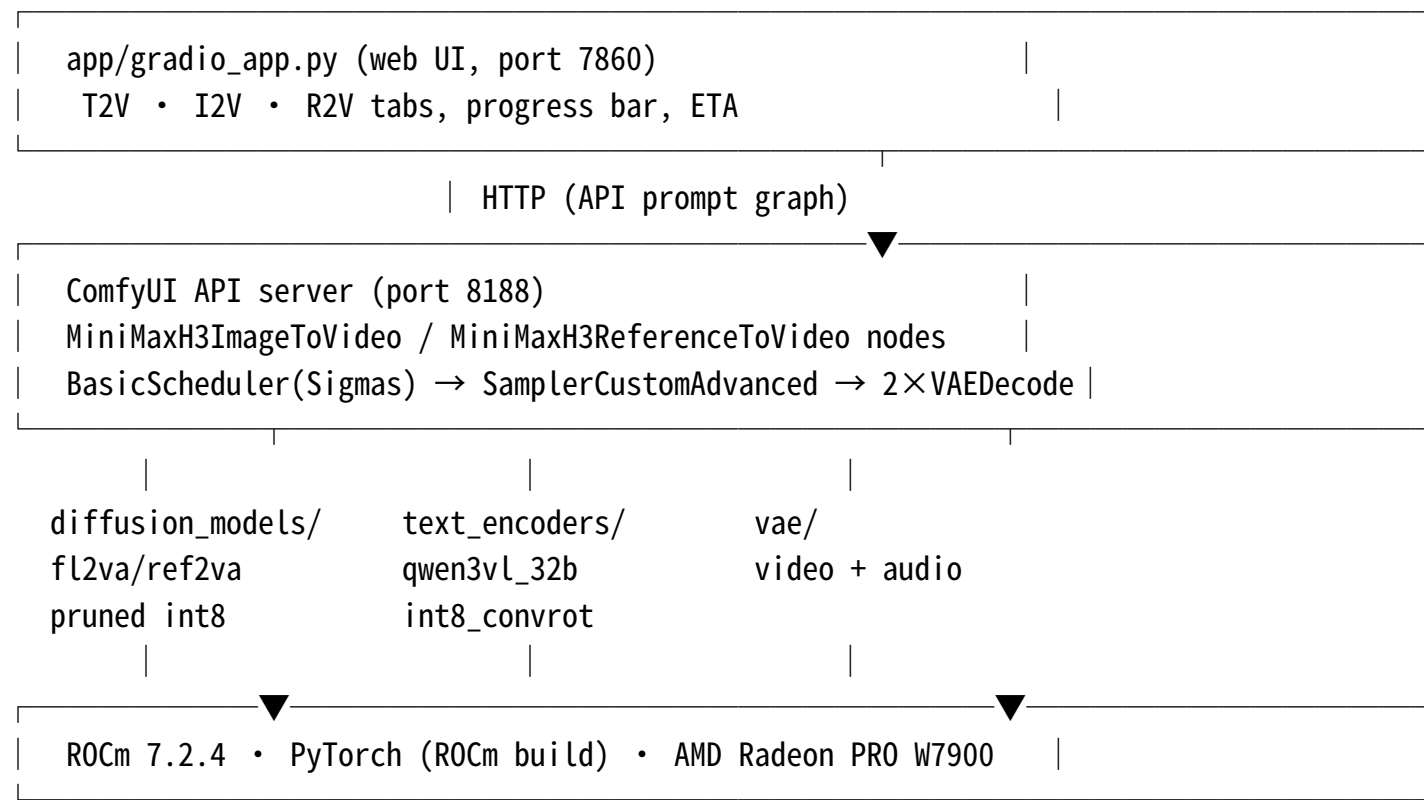
However, the open-weights release is enormous (bf16 models exceed 66 GB for the diffusion backbone alone plus a 51 GB 32B text encoder), and most inference frameworks assume NVIDIA hardware. This project adapts MiniMax H3 to run **entirely on an AMD Radeon GPU via ROCm**, using a carefully chosen **pruned + int8** weight set and ComfyUI's native nodes, and packages it as a simple local web studio.

## 2. Target Users & Application Scenarios

| User group             | Scenario                                                        | Value                                    |
|------------------------|-----------------------------------------------------------------|------------------------------------------|
| Individual creators    | Short-form video, social clips with synchronized sound          | Free local generation, no cloud fees     |
| Agencies / advertisers | Product ads, storyboards, reference-style character reuse (R2V) | Private data stays on-device             |
| E-learning teams       | Explainers, animated diagrams, localized voiceover audio        | Audio generated in-loop, no TTS pipeline |
| IP / game studios      | Cinematics, style-locked shots (R2V identity lock)              | Rapid iteration on reference frames      |
| Privacy-sensitive orgs | Medical / financial / legal visualization                       | 100% local inference, no uploads         |

The tool is deliberately a **single-machine, single-GPU** studio: install once, start two processes, and generate.

### 3. System Architecture



- **Client** (`comfy_client.py`): builds ComfyUI API prompt graphs for the three modes, uploads keyframe/reference images, polls history, returns the mp4.
- **Server** (ComfyUI): runs MiniMax H3's native nodes; the omni-modal sampler decodes **video latents and audio latents in the same sampling loop**, then the two VAEs render frames + stereo audio into a single mp4.
- **UI** (`gradio_app.py`): zero-setup web studio.

### 4. Model & Algorithm Introduction

#### 4.1 MiniMax H3

MiniMax H3 is an omni-modal generative model that produces video and audio from text/image conditioning. Key traits:

- **Unified video + audio generation** — one sampling pass renders both, so lip-sync/ambience/score are naturally aligned (no post-hoc dubbing).
- **T2V / I2V / R2V modes** — text-only, keyframe-constrained, or identity/style locked from reference images (`<Picture 1>` reference syntax).

- **Length grid** — frames snap to a 17k+5 grid at 24 fps (~5 s = 124 frames; trained range roughly 124–362).

## 4.2 Adaptation choices for AMD Radeon / ROCm

| Choice                                                      | Why                                                                                                           |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>pruned int8 diffusion</b><br>(fl2va/ref2va, ~21 GB each) | bf16 is 66 GB — would not fit 48 GB VRAM alongside the text encoder; pruned set removes training-only weights |
| <b>int8 Qwen3-VL-32B text encoder</b> (~27 GB)              | int8 (convrot) runs natively on ROCm; nvfp4-awq would need FP4 emulation on RDNA                              |
| <b>fp16 video VAE + fp32 audio VAE</b>                      | small, full precision where it matters                                                                        |
| <b>ComfyUI native nodes</b>                                 | no custom CUDA kernels; the same nodes run on ROCm PyTorch                                                    |
| <b>20 steps · res_multistep · simple</b>                    | Comfy-Org reference config, good quality/speed balance                                                        |

## 5. Adaptation Description for AMD Radeon GPU / ROCm

The entire inference stack runs on AMD hardware:

- **ROCm 7.2.4** runtime with **gfx1100** (RDNA3) support — our AMD Radeon PRO W7900 is detected natively (rocm-smi: DID 0x744b, RDNA3).
- **PyTorch ROCm build** (torch 2.10.0+rocm7.2.4) drives ComfyUI's HIP kernels — no CUDA translation layers, no NVIDIA-only code paths.
- **Measured end-to-end** on the W7900 (48 GB, 864 GB/s):

| Run             | Resolution | Duration | Wall time                             |
|-----------------|------------|----------|---------------------------------------|
| T2V cold start  | 864x480    | ~5.2 s   | 10:42 (model load included)           |
| T2V warm        | 608x352    | ~2.3 s   | 75 s                                  |
| T2V warm        | 864x480    | ~2.3 s   | 170 s                                 |
| <b>I2V warm</b> | 864x480    | ~5.2 s   | <b>~10 min</b> (keyframe-conditioned) |

Peak VRAM ~27 GB (UNet) — comfortably inside 48 GB even with the 27 GB text encoder resident.

**Honest performance disclosure:** image/reference-conditioned modes (I2V/R2V) are the slowest — conditioning tokens ride through

every sampling step. A ~5 s I2V clip at 864x480 takes ~9-10 min on this card (553–617 s measured). We report this openly rather than only showing best-case T2V numbers.

## 6. Innovation Highlights

1. **Full open-weights omni-modal studio on a single AMD GPU** — video and audio from one sampling loop, no cloud calls, no external TTS.
2. **Pruned + int8 quantization engineering** — the practical recipe (which files to pick from the official repack, why each) that makes a 190 GB+ bf16 release fit a 48 GB card.
3. **Three modes in one API client** — T2V / I2V / R2V share one prompt-graph builder; reference images ride the sampling loop for identity lock.
4. **ROCm-first setup** — PyTorch ROCm + ComfyUI native nodes only; reproducible on any RDNA2/RDNA3 card with  $\geq 24$  GB.

## 7. Reproducibility

# 1. ComfyUI

```
git clone https://github.com/comfyanonymous/ComfyUI.git /opt/ComfyUI
/opt/venv/bin/pip install -r /opt/ComfyUI/requirements.txt
```

# 2. models (HF mirror, ~75 GB)

# see README §Quick start for exact hf download commands

# 3. ComfyUI server

```
cd /opt/ComfyUI && /opt/venv/bin/python main.py --listen 0.0.0.0 --port 8188
```

# 4. web studio

```
cd minimax-h3-tool && /opt/venv/bin/python app/gradio_app.py --port 7860
```

See [README.md](#) for the full dependency list, and [benchmarks/results.md](#) for performance reproduction.

## 8. Demo Video

3–5 minute walkthrough showing: environment (rocm-smi + ComfyUI boot) → T2V generation from the web UI → output video with audio → (I2V/R2V if shown). Generated clips are included in the submission.

## 9. Legal & Data

- Model license: **minimax-h3-community-license-agreement** (non-commercial community license).
- All generation is local; no user data leaves the machine.